A glycaemic digital twin that warns of highs and lows two hours ahead

Type 2 diabetes · EHR + wearables · Happiest Health Digital Twin Challenge 2026

Team name · College / incubator name

Synthetic-data research prototype. Not for clinical use.

Highs and lows happen between clinic visits

101 M

people with diabetes in India, mostly Type 2 (ICMR-INDIAB, Lancet Diabetes & Endocrinology 2023)

3 months

between HbA1c tests, while dangerous highs and lows happen within hours

At night

lows from sulfonylureas or insulin strike while the patient is asleep and cannot react

The gap: the record (diagnoses, labs, drugs) and the sensor (glucose every 5 minutes) live in separate systems. Nobody combines them into an early warning a doctor can act on.

Our answer: a digital twin of each patient

1

Predict

Risk of glucose > 180 or < 70 mg/dL in the next 2 hours, every 30 minutes, from 6 h of wearable data plus the EHR

2

Alert

One alert per risk episode, tuned so clinicians are not flooded: about one low-glucose alert every two days

3

Explain

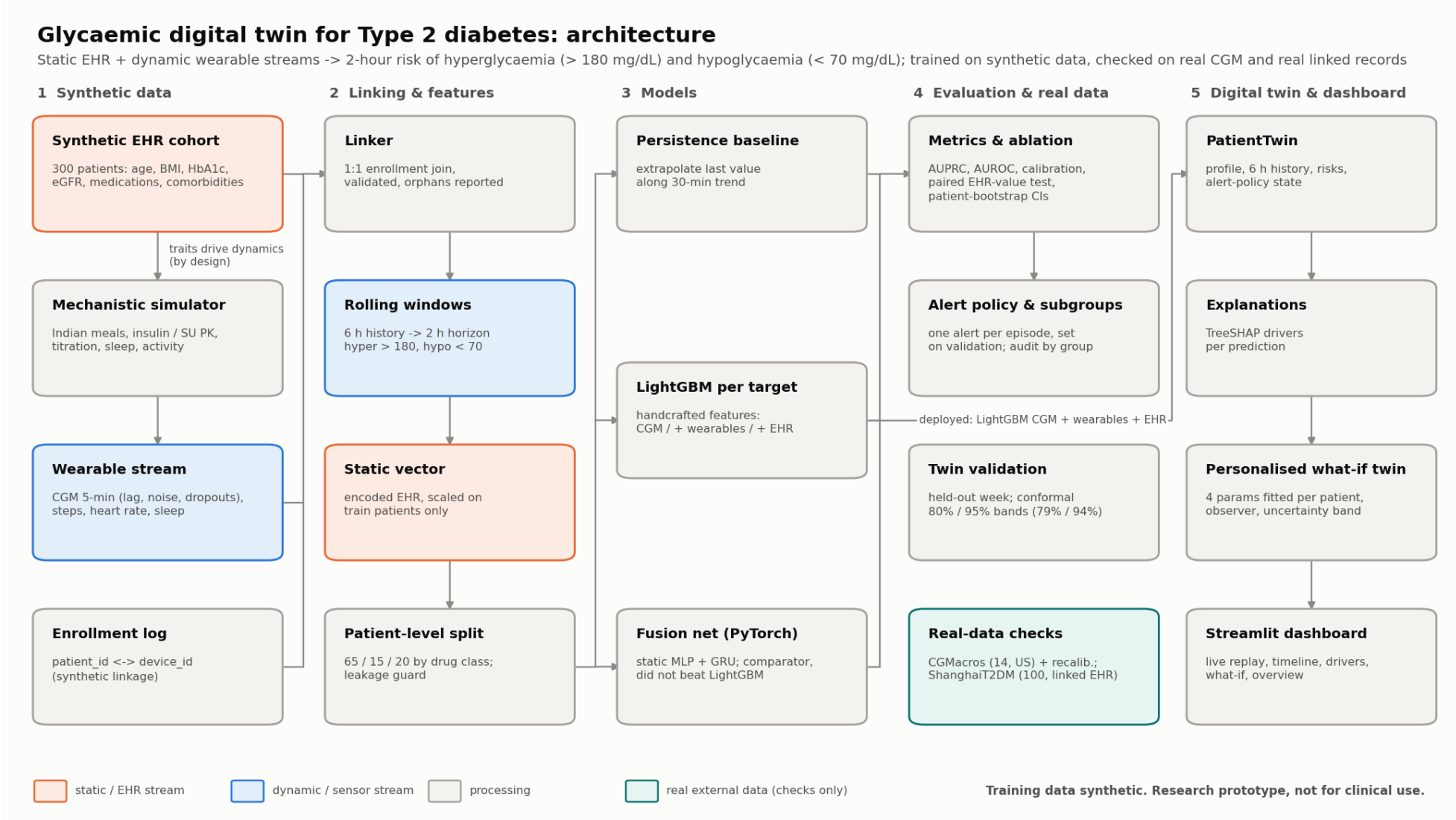
The top reasons behind every risk (TreeSHAP), e.g. falling glucose, sulfonylurea dose, time of day

4

Simulate

A twin fitted to the patient's own data answers 'what if?' (late dinner, skipped meal, extra dose) with an uncertainty band

Architecture: five layers from data to dashboard



Data: a physiology simulator, honestly labelled

What we generate

- 300 patients × 14 days, 5-minute CGM with sensor lag, noise and dropouts
- EHR: age, BMI, HbA1c, eGFR, lipids, comorbidities, metformin / sulfonylurea / insulin doses
- Physiology: Bergman-style glucose model, Indian meals (rice, roti, idli, poha), drug pharmacokinetics, sleep, activity, dawn effect
- Patient-level split 194 / 46 / 60, stratified by drug class

Why synthetic, and what that means

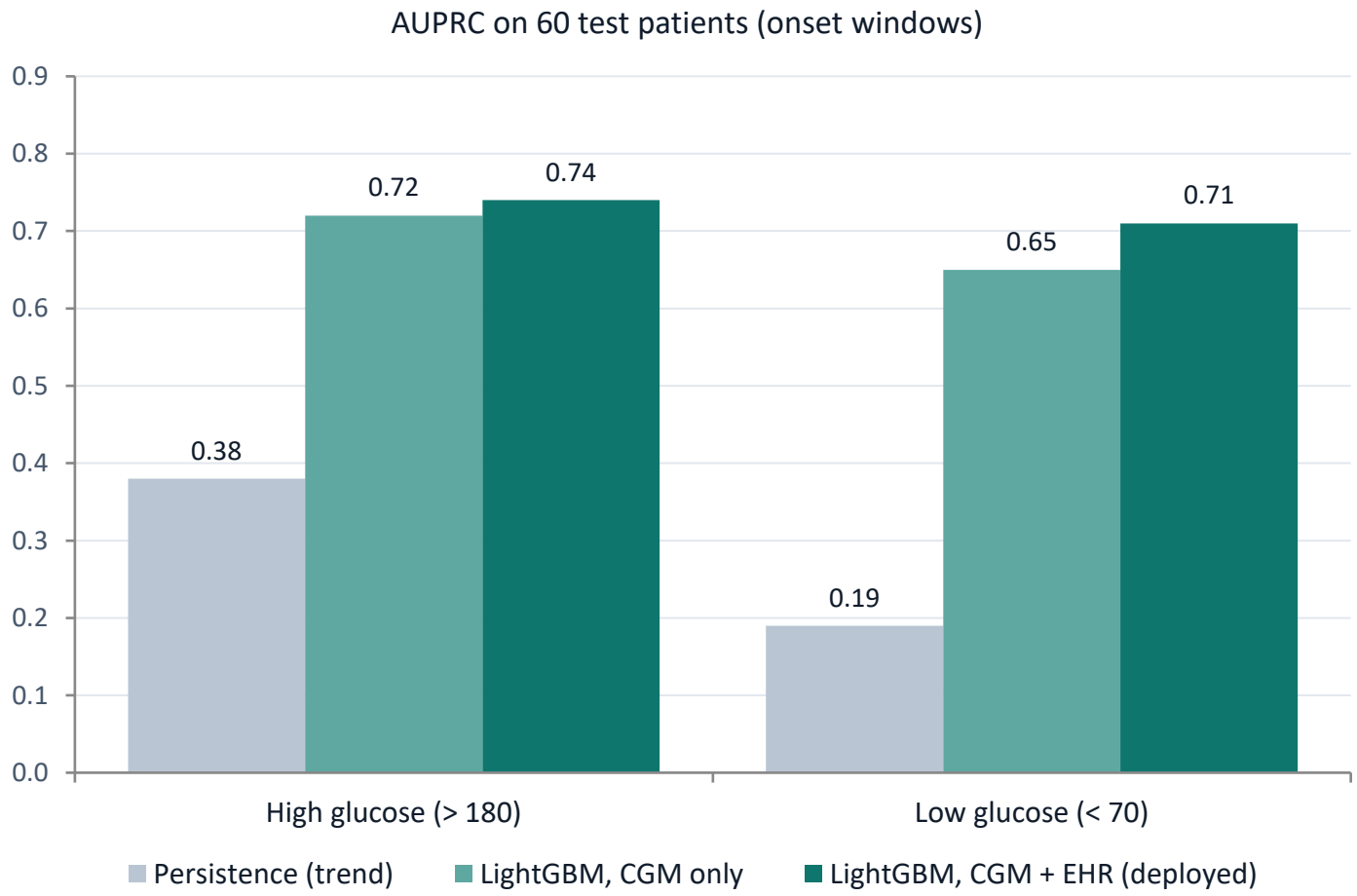
No public dataset links hospital records and wearables for the same Indian patients, so we simulate both.

The link is synthetic by design, and we say so everywhere. The simulator makes EHR traits drive glucose, so any EHR gain on this data is partly built in. That is why we went on to test the EHR on real linked records.

Built so the numbers can be trusted

1	**Patient-level split**	No patient appears in both training and test; the code raises an error if one does
2	**Onset windows**	We score predictions made before the event starts, not once glucose is already low
3	**Paired patient bootstrap**	Every comparison comes with a 95% confidence interval over patients
4	**Choices made on validation**	Alert thresholds and the alert policy never see the test patients
5	**Reproducible**	Fixed seeds, one command rebuilds everything, 128 automated tests

On synthetic patients the model predicts both events well



AUROC 0.93 for highs, 0.99 for lows

AUPRC is the honest metric for rare lows: chance is 0.02

- A GRU fusion network did not beat LightGBM
- Steps, heart rate and sleep added nothing measurable
- Well calibrated (hypo ECE 0.004), so we deploy LightGBM
- These numbers are optimistic: simulated lows drift in slowly

Alerts a clinic can live with

76%

of low-glucose episodes warned at least 30 minutes ahead

65 min

median warning time before a low begins

0.49

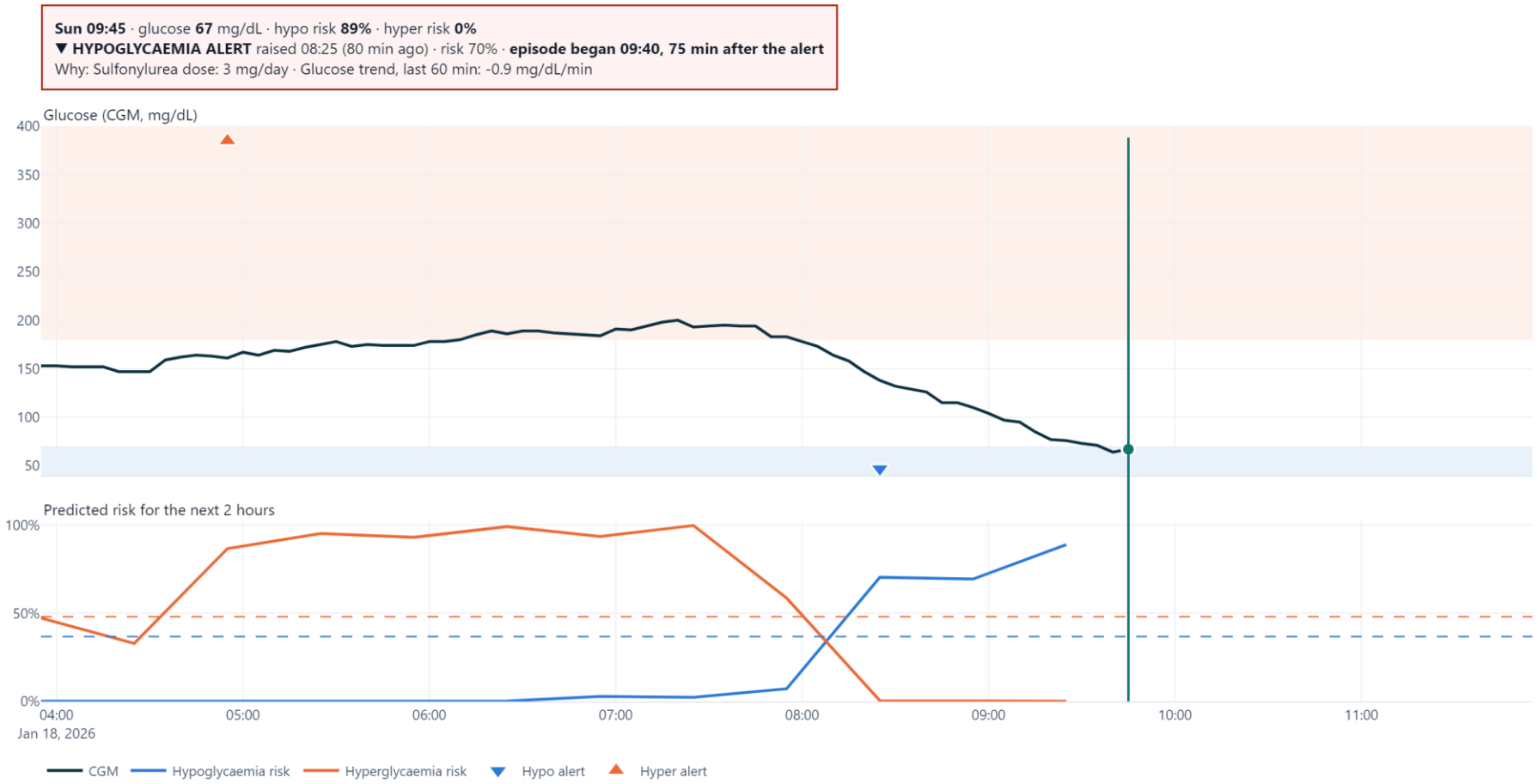
low-glucose alerts per patient per day (about one every two days)

66%

of low-glucose alerts are followed by a real low

Why a policy matters: raising an alert every time the risk is high would mean about 3 alerting windows per patient-day with only 18% true. Our policy alerts once per risk episode, never while the CGM already reads low, and was tuned on validation patients only. It trades a little sensitivity (98% to 76% caught 30 min ahead) for 6x fewer alerts.

Live replay: the alert arrives 75 minutes before the low



08:25

Hypo alert raised at 70% risk

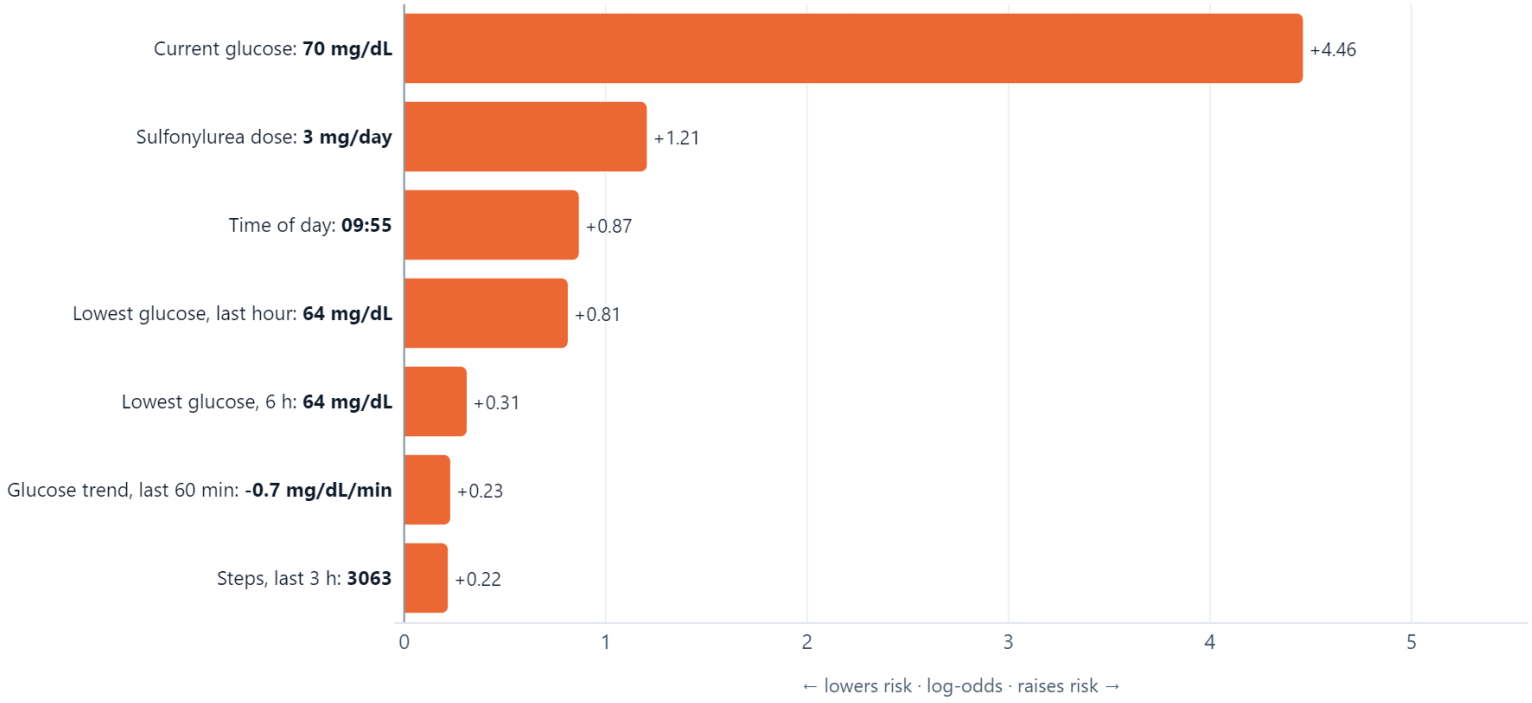
Why

sulfonylurea dose, glucose falling 0.9 mg/dL per min

09:40

Glucose falls below 70

Every alert explains itself



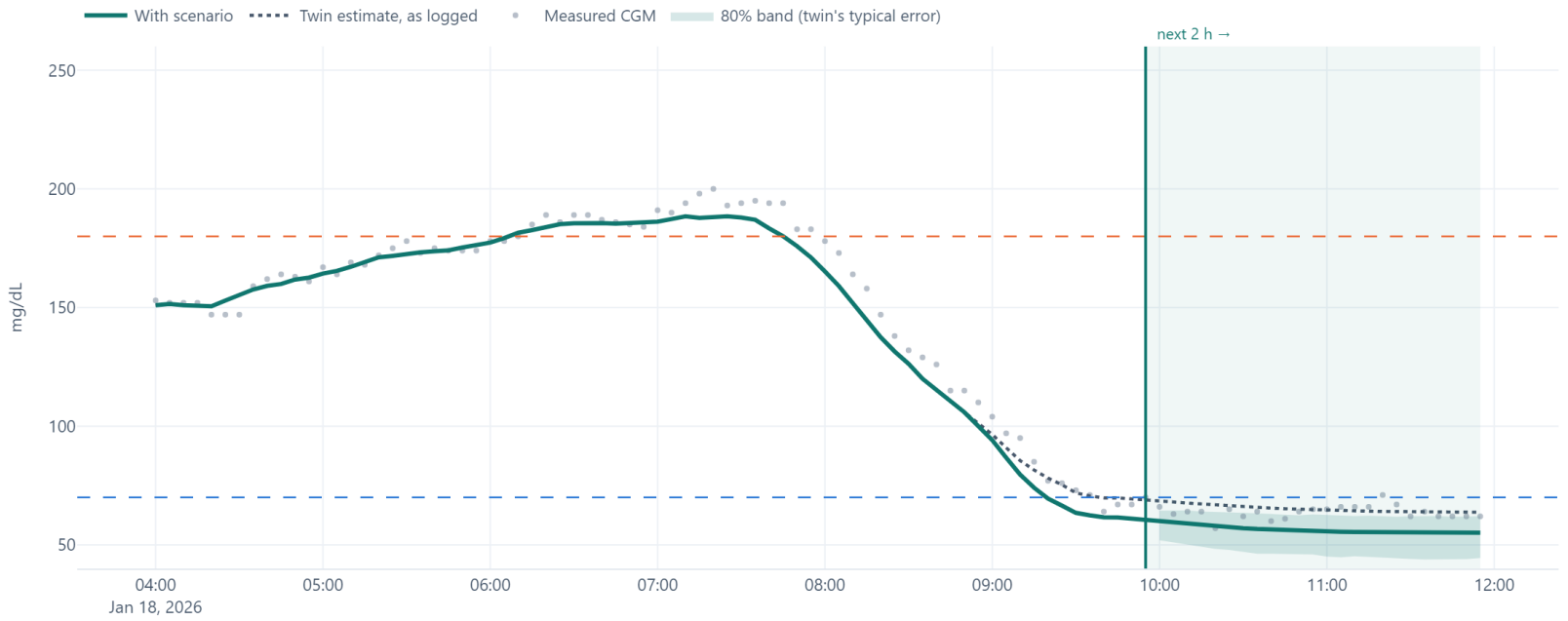
Example: hypoglycaemia risk 95% for patient P0014 (synthetic). Current glucose, the sulfonylurea dose and the time of day push the risk up.

TreeSHAP drivers

Exact contribution of each input to this patient's risk at this moment

- Grouped by data stream: CGM, time, activity and sleep, EHR
- Written as plain sentences for the clinician
- They explain the model, not the physiology, and the dashboard says so

A personalised twin answers 'what if?'



How it works: 4 physiology parameters fitted to the patient's first week (basal glucose, carb sensitivity, clearance, drug sensitivity); a state observer tracks the CGM up to now; scenarios are re-simulated with a calibrated uncertainty band. Shown: an extra 8 U insulin dose pushes the projected low from 64 to 55 mg/dL.

5.9 vs 12.1

mg/dL error on a held-out week: fitted twin vs EHR defaults

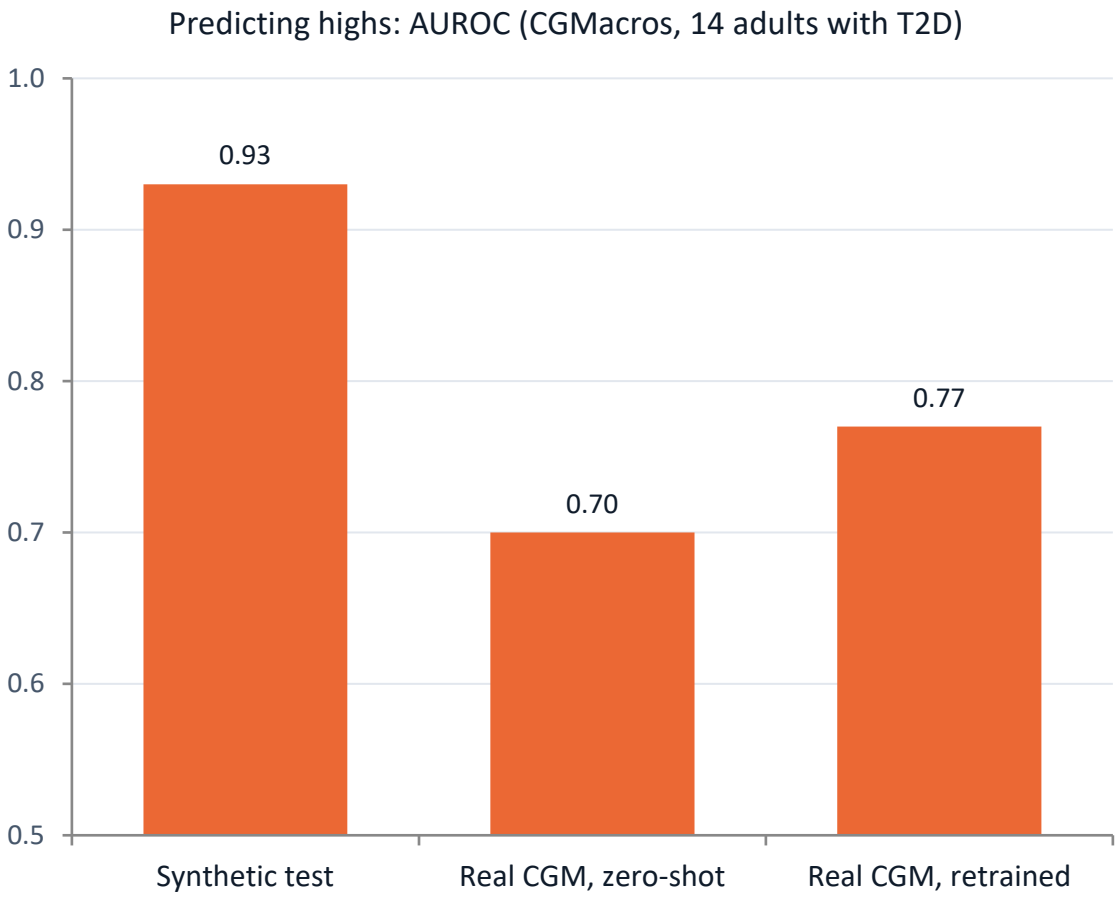
79%

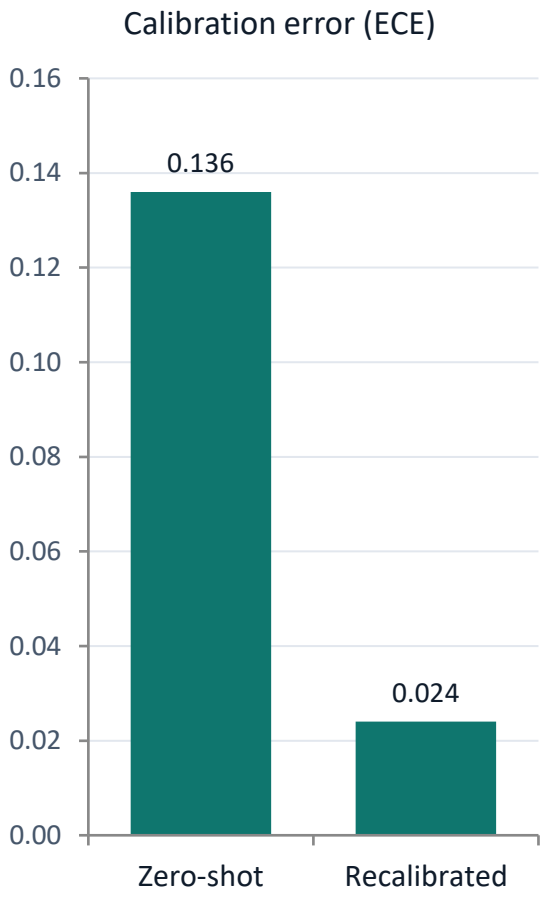
of real readings fell inside the 80% band on unseen test patients

Then we tested it on real people

Two public datasets, never used for training: 14 US adults (CGMacros) and 100 Chinese patients whose CGM comes with their own clinical records (ShanghaiT2DM)

Real CGM: performance drops, and we show it

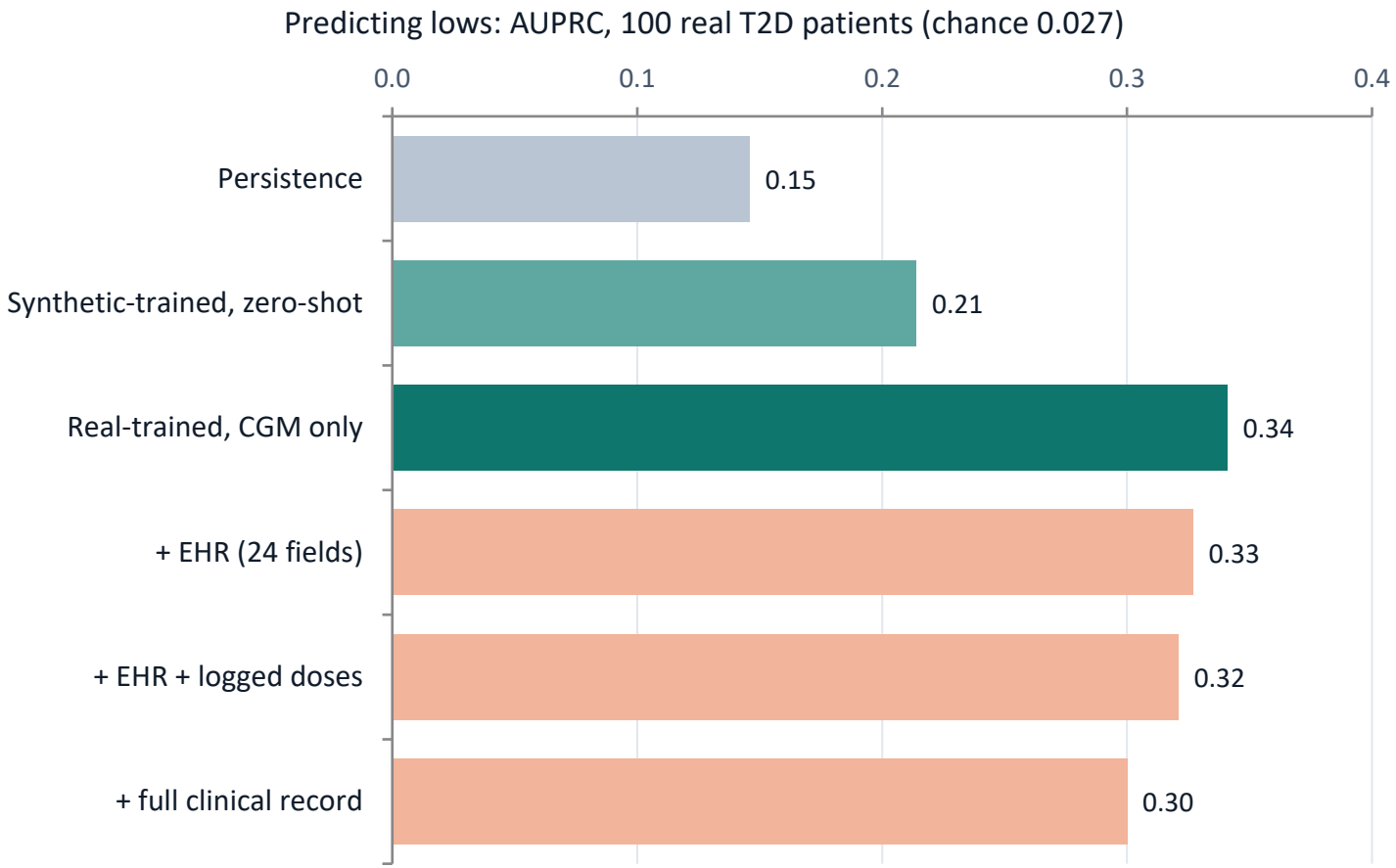




Honest gap
Simulated glucose is smoother than real glucose

Fixable
Recalibrating on other real patients makes the probabilities trustworthy without retraining

Real linked records: the CGM model predicts real lows well

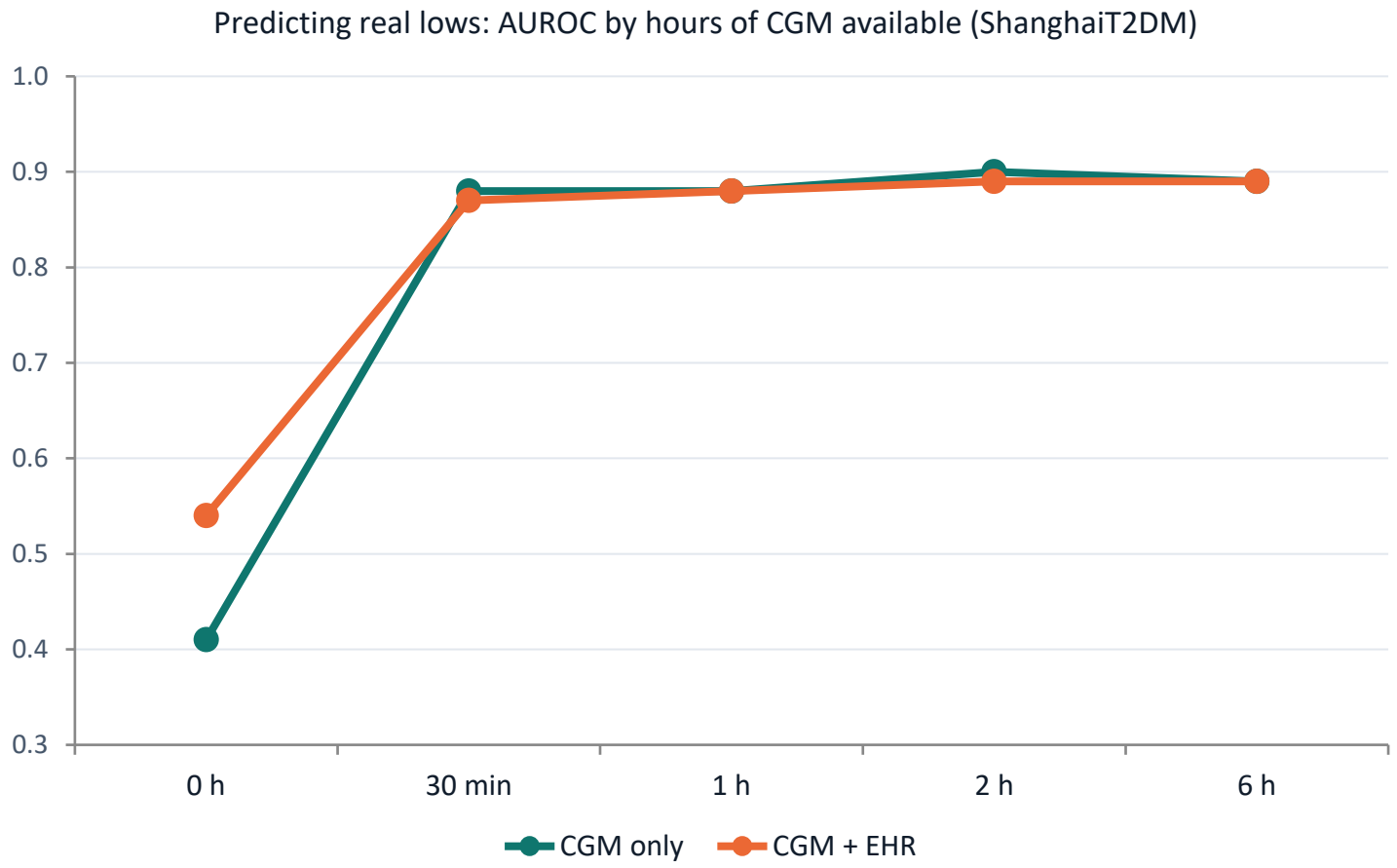


0.89 AUROC for real lows (nocturnal 0.91), well calibrated

+0.07 synthetic-trained model beats trend extrapolation on real lows

No gain from the EHR, logged doses or the full record

The EHR helps only before the sensor starts



No CGM yet
The EHR lifts AUROC from chance to 0.54: a weak but real signal for deciding who to watch

30 minutes of CGM
already reaches 0.88, almost as good as 6 hours. From then on the EHR adds nothing

So the EHR is a prior, not a forecaster

What we claim, and what we no longer claim

We claim

- A CGM-based twin predicts real lows 2 hours ahead (AUROC 0.89 on 100 real patients)
- Alerting can be tuned to about one alert every two days
- Recalibration on local data fixes the probabilities
- A personalised twin halves projection error, with honest bands

We no longer claim

- That adding the EHR improves the 2-hour forecast: true on synthetic data only
- That synthetic accuracy (0.93) is what real patients would see
- Any clinical validity: this is a research prototype

Path to real use in India

1a

Real-data study

Real linked EHR + CGM from Indian T2D clinics, ethics approval (ICMR AI guidelines)

1b

Recalibrate

Fit probabilities and alert thresholds to the local population

1c

Silent pilot

Model scores in the background; outcomes compared afterwards

1d

Clinician pilot

Diabetologists judge usefulness, workload and safety

DPDP Act 2023 & Rules 2025: consent, purpose limits, security

ABDM / ABHA consent flows for linking record and device

CDSCO medical device software rules (risk class to confirm)

CGM alarms stay on; the twin never replaces them

Built to be checked, not just demoed

128
automated tests

1
command rebuilds data, models and reports

2
public real datasets checked

Live
dashboard on Streamlit Cloud

Open and documented: MIT licence, model card, deployment path, data licences (CC BY 4.0, CC BY-NC-SA 4.0), architecture diagram and every result table in the repository. Real datasets are downloaded by the user and never committed.

Python · LightGBM · PyTorch · SHAP · SciPy · scikit-learn · Plotly · Streamlit

Limitations we know about

Synthetic training data

The deployed model has never been trained on real patients

Small real checks

14 US and 100 Chinese patients, not Indian outpatients

Insulin users

Their lows are the hardest to predict, on synthetic and real data

Inputs a real twin needs

Planned meals and fresh doses are invisible to the model until glucose moves

A twin that warns early, explains itself, and is honest about what it knows

Next: validate on real linked records from Indian clinics, with clinicians in the loop

github.com/AkhileshNalla11/glycaemic-digital-twin

Live demo and video links: see the README

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